

2) Sinking $\rightarrow a \neq 0 \downarrow$

$$W_0 > F_b$$

$$m_0 > m_f$$

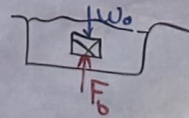
$$\rho_0 V_0 > \rho_f V_f$$

$$\rho_0 > \rho_f$$

$$a = \frac{m_0 - m_f}{m_0} \times g \text{ or } a = \frac{\rho_0 - \rho_f}{\rho_0} \times g$$

fully immersed $\rightarrow V_0 = V_f$

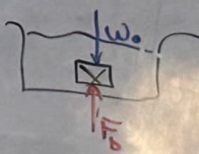
3) Moving up $\rightarrow a \neq 0 \uparrow$



$$\rho_f > \rho_0$$

$$a = \frac{m_f - m_0}{m_0} \times g = \frac{\rho_f - \rho_0}{\rho_0} \times g$$

4) rest $\rightarrow a = 0$



$$W_0 = F_b$$

$$m_0 = m_f$$

$$\rho_0 V_0 = \rho_f V_f$$

$$\rho_0 = \rho_f$$

Apparent weight

The weight of the object in air is more than its weight in fluid

$$W_{\text{fluid}} = W_{\text{air}} - F_b$$

$$F_b = W_{\text{air}} - W_{\text{fluid}}$$

- ① 50 N object displaces 10 N fluid
- ② 50 N object, its weight reduced to 30 N
- ③ 50 N object, its weight reduced by 20 N

$$\begin{aligned} \rightarrow W_{\text{air}} &= 50 \text{ N} \\ \rightarrow W_{\text{fluid}} &= 40 \text{ N} \\ \rightarrow F_b &= 10 \text{ N} \end{aligned}$$

$$\begin{aligned} \rightarrow W_{\text{air}} &= 50 \text{ N} \\ \rightarrow W_{\text{fluid}} &= 30 \text{ N} \\ \rightarrow F_b &= 20 \text{ N} \end{aligned}$$

$$\begin{aligned} \rightarrow W_{\text{air}} &= 50 \\ \rightarrow W_{\text{fluid}} &= 20 \text{ N} \\ \rightarrow F_b &= 30 \text{ N} \end{aligned}$$

Apparent weight

ex. 6

6cm object having $\rho_o = 800 \text{ kg/m}^3$
floats in fluid having $\rho_f = 1200$

a) The depth immersed is:

$$\rho_o h_o = \rho_f h_f \rightarrow h_f = \frac{\rho_o}{\rho_f} \times h_o = \frac{800}{1200} \times 6 = 4 \text{ cm}$$

b) if the object is held immersed
and then released. The acceleration is:

$$a = \frac{\rho_f - \rho_o}{\rho_o} \times g = \frac{1200 - 800}{800} \times 9.8 = 4.9$$

ex. 7

$9.5 \times 10^2 \text{ N}$ metal. if a metal is immersed
in water its weight reduced to $7 \times 10^2 \text{ N}$

The density of the metal is:

$$\left. \begin{array}{l} W_{\text{air}} = 9.5 \times 10^2 \text{ N} \\ W_{\text{fluid}} = 7 \times 10^2 \text{ N} \end{array} \right\} F_b = 2.5 \times 10^2 \text{ N}$$

$$F_b = W_{\text{air}} \times \frac{\rho_f}{\rho_o}$$

$$\rho_o = \frac{W_{\text{air}}}{F_b} \times \rho_f = \frac{9.5 \times 10^2}{2.5 \times 10^2} \times 1000 = 3800 \text{ kg/m}^3$$

$$F_b = \frac{1000}{\rho_f} \rho_o g$$

ex. 8

50 N object floats in water
if the depth immersed is 3cm
and above a water is 1cm

Find the volume of the object

$$h_o = 3 + 1 = 4 \text{ cm}$$

$$h_f = 3 \text{ cm}$$

$$\rho_o h_o = \rho_f h_f$$

$$\rho_o = \rho_f \frac{h_f}{h_o} = 1000 \times \frac{3}{4} = 750$$

$$W_o = \rho_o V g$$

$$V_o = \frac{W_o}{\rho_o g} = \frac{50}{750 \times 10} = 0.0067 \text{ m}^3$$